

Yao Ling

+86 17314959714 | zero-in-ur-area@sjtu.edu.cn

Education

Shanghai Jiao Tong University (SJTU)

Bachelor of Engineering in IntelliSense

Shanghai, China

Sep. 2023 – June 2027

KTH Royal Institute of Technology

Exchange Student in Computer Science

Stockholm, Sweden

Aug. 2025 – Jan. 2026

- **Graduate-level coursework:** Introduction to Robotics (A), Applied Estimation (A), Digital Signal Processing (A), Artificial Intelligence (A)

Research Projects

Distributed Optimization via Chebyshev Polynomial Approximation

Feb. 2025 – Present

Role: Research Assistant at **IWIN-FINS Lab** (Advisor: **Prof. Jianping He**); Supported by **Chun-Tsung Program**

Problem: Addressed slow convergence and high communication complexity in large-scale multi-robot networks lacking central authority (e.g. underwater robots, sensor networks).

Methodology: Proposed a novel **Chebyshev polynomial-based surrogate algorithm** to approximate local objectives. This approach replaces raw gradient transmission with polynomial coefficients, significantly reducing communication rounds while theoretically preserving convergence guarantees and bounding approximation errors in non-convex landscapes.

State Estimation Paradigms: Robustness vs. Efficiency in UAV Navigation

Nov. 2025 – Jan. 2026

Role: Independent Researcher

Problem: Addressed the fundamental "No Free Lunch" dilemma in UAV localization: **EKF** lacks global recovery ("Kidnapped" scenarios), while **PF** suffers from high computational cost and "jitter" in low-noise regimes.

Methodology: Designed a deterministic simulation framework in **Unity** to conduct controlled stress testing across distinct scenarios:

- ◇ *Linearization Limits:* Tested EKF failure modes under abrupt state discontinuities.
- ◇ *Global Recovery:* Evaluated **Augmented MCL**'s ability to recover from "kidnapping" within 2 seconds.
- ◇ *Sample Efficiency:* Investigated a **Hybrid EKF-PF architecture** to reduce particle dependency.

Experience

Teaching Assistant

National Mathematical Modeling Training Camp, SJTU

Jun.2025 – Aug. 2025

Shanghai, China

- Awarded **Outstanding Teaching Assistant** (Top tier) for mentoring students in modeling and report writing.

Robotics Engineer Intern

Tohoku University Engineering Summer Program (JASSO Scholar)

July 2024 – Aug. 2024

Sendai, Japan

- Engineered an autonomous rover using **ROS**, deploying **YOLOv8** for obstacle detection and **A*** for path planning.

Honors & Awards

First Prize, RoboMaster Robotics Competition (Hardware & Control)

2023

Honorable Mention, Mathematical Contest in Modeling (MCM), COMAP

2024

JASSO Scholarship, Japan Student Services Organization

2024

Chun-Tsung Scholar, SJTU Undergraduate Research Fellowship (Top 1%)

2025

Academic Excellence Scholarship, Shanghai Jiao Tong University

2025

Technical Skills

English: TOEFL: 104, CET6: 607

Programming: C++, Python, MATLAB, C#, C, Node.js

Robotics & Control: ROS, State Estimation, Motion Planning, PID Control

Vision & AI: PyTorch, OpenCV, YOLO, Diffusion Models